

Lesson 10: Applications of Slope Fields (Qualitative Curve Tracing & Asymptotes)

Professor Leonard Differential Equations Master Series · Lesson 10

- **Direct Video Link:** [Lesson 10: Applications of Slope Fields \(Qualitative Curve Tracing & Asymptotes\)](#) · Duration: 38:42
- **Target Exam Scope:** 🎯 MIDTERM EXAM (Ch 1.2 — Qualitative Asymptotic Behavior)
- **Course Context:** Applied Ordinary Differential Equations (ENGR 213)

1. Whiteboard Lecture Intuition & Philosophical Framing

In this 39-minute lecture, Professor Leonard demonstrates how to deduce the long-term qualitative behavior of solutions purely from slope fields. Without evaluating complicated integrals, engineers can predict limits as $x \rightarrow \infty$, identify attractor/repeller asymptotes, and guarantee stability.

2. Core Theoretical Concepts & Mathematical Formulations

Qualitative Analysis Without Solving

Often, non-linear ODEs cannot be integrated in terms of elementary functions. Qualitative analysis answers the critical engineering questions:

1. Does the solution blow up to $\pm\infty$ in finite time?
2. Does the solution approach a steady-state equilibrium?
3. How sensitive is the final state to initial disturbances?

Asymptotic Attractors vs. Repellers

- **Attractor (Stable Asymptote):** Solution curves that start near the asymptote are channeled into it as $x \rightarrow \infty$.
 - **Repeller (Unstable Asymptote):** Solution curves diverge away from the curve as time progresses.
 - **Funneling Theorem:** If two bounding trajectories converge, all intermediate solutions are trapped between them for all future time.
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3. Classroom Chalkboard Examples (Solved in Complete Baby Steps)

Problem 10.1: Chalkboard Problem 10.1: Predicting Asymptotic Limits from Slope Field Analysis

Problem Statement:

Analyze the autonomous differential equation $\frac{dy}{dx} = y(2 - y)$.
1. Find all equilibrium solutions (where $dy/dx = 0$).
2. Determine the sign of dy/dx across the regions separated by the equilibrium lines.
3. Sketch the qualitative trajectories for initial values $y(0) = -1$, $y(0) = 1$, and $y(0) = 3$.
4. State $\lim_{x \rightarrow \infty} y(x)$ for each initial condition.

Step-by-Step Whiteboard Solution:

- **Step 1: Find Equilibrium Solutions:** Set $\frac{dy}{dx} = 0 \implies y(2 - y) = 0 \implies y = 0$ and $y = 2$. These represent two horizontal straight-line solutions that partition the plane into 3 horizontal zones.
- **Step 2: Sign Analysis Across Zones:**
 - **Zone 1 ($y > 2$):** Pick test point $y = 3$: $dy/dx = 3(2 - 3) = -3 < 0$. Slopes are negative; curves decrease toward $y = 2$.
 - **Zone 2 ($0 < y < 2$):** Pick test point $y = 1$: $dy/dx = 1(2 - 1) = +1 > 0$. Slopes are positive; curves increase toward $y = 2$.
 - **Zone 3 ($y < 0$):** Pick test point $y = -1$: $dy/dx = -1(2 - (-1)) = -3 < 0$. Slopes are negative; curves plunge toward $-\infty$.
- **Step 3: Evaluate Asymptotic Limits for Given Initial Conditions:**
 - For $y(0) = 3$ (in Zone 1): Since $dy/dx < 0$ and bounded below by $y = 2$:
$$\lim_{x \rightarrow \infty} y(x) = 2$$
 - For $y(0) = 1$ (in Zone 2): Since $dy/dx > 0$ and bounded above by $y = 2$:
$$\lim_{x \rightarrow \infty} y(x) = 2$$
 - For $y(0) = -1$ (in Zone 3): Since $dy/dx < 0$ and accelerating downwards:
$$\lim_{x \rightarrow \infty} y(x) = -\infty$$
- **Conclusion:** $y = 2$ is an **asymptotically stable equilibrium (attractor / sink)**, while $y = 0$ is an **unstable equilibrium (repeller / source)**.

[!WARNING] **Common Exam Pitfall:** Failing to recognize that distinct solution curves cannot intersect each other (by Picard's Uniqueness Theorem). A curve starting in Zone 2 can never cross the equilibrium line $y = 2$.

or $y = 0$!

4. Key Takeaways & Exam Strategy Formula Card

- Watch the full whiteboard demonstration on YouTube: [Lesson 10: Applications of Slope Fields \(Qualitative Curve Tracing & Asymptotes\)](#).
- Master the classification and verification mechanics before attempting complex integrations.
- Always check domain restrictions and verify initial values back into the computed trajectory.